

UNIT VII: NETWORK SECURITY(5 HRS)

7.1. Overview of Network Security

7.2. Email Security: S/MIME, Pretty Good Privacy (PGP)

7.3. Secure Socket Layer (SSL) and Transport Layer Security (TLS)

7.4. IP Security (IPSec)

7.5. Firewalls and their types

OVERVIEW OF NETWORK SECURITY

- Network security is the practice of protecting computer networks and the data they carry from unauthorized access, misuse, modification, or disruption.
- Network security refers to the measures taken by an enterprise or organization to secure its computer networks and data using both **hardware and software solutions**.
- The main objective is to ensure the **confidentiality, integrity, and availability** of data and network resources.
- In today's digital world, every organization that handles large amounts of sensitive data must implement strong network security solutions to defend against various cyber threats such as hacking, malware, ransomware, and data breaches.

WHY NETWORK SECURITY MATTERS

- Networks are the backbone of modern organizations and the internet.
- Billions of devices are connected to a network (IoT, cloud, mobile).
- Attacks are increasing in frequency, scale, and sophistication (ransomware, state-sponsored attacks, data breaches).
- A single breach can cause financial loss, reputational damage, legal penalties, and loss of customer trust.
- **Core Objective:** Implement **Defense-in-Depth** — multiple layers of security so that if one layer fails, others still protect the system.

PRIMARY GOALS OF NETWORK SECURITY (CIA TRIAD)

The core objectives of network security are:

- **Confidentiality**

- Ensures that sensitive information is accessible **only to authorized persons**.
- Protects business data from unauthorized viewing or disclosure.
- Achieved through encryption, access control, and strong authentication.

- **Integrity**

- Maintains the **accuracy, consistency, and trustworthiness** of data.
- Ensures data is not altered or tampered with by unauthorized users during storage or transmission.
- Achieved using hashing, digital signatures, and checksums.

- **Availability**

- Ensures that data and network resources/services are **accessible and usable** by legitimate users whenever needed.
- Protects against attacks that try to make services unavailable (e.g., DDoS attacks).
- Achieved through redundancy, backups, failover systems, and DDoS mitigation.

TYPES / LEVELS OF NETWORK SECURITY

Network security is implemented at three major levels:

- **Physical Network Security**

- The most basic and fundamental level.
- Protects the physical components of the network (servers, routers, switches, cables, data centers).
- Prevents unauthorized physical access.
- **Measures:** Biometric locks, CCTV, security guards, locked server rooms, tamper-proof hardware, and restricted access to network devices.

- **Technical Network Security**

- Focuses on protecting data stored in the network and data in transit.
- Protects against both **unauthorized users** and **malicious software/attacks**.
- **Tools & Technologies:** Firewalls, Antivirus/Anti-malware, Encryption (TLS, IPSec), IDS/IPS, VPNs, secure protocols, etc.

- **Administrative Network Security**

- Also known as **Procedural or Policy-based security**.
- Deals with **user behavior, policies, and procedures**.
- Covers how permissions and authorizations are granted and managed.
- Involves creating security policies, user training, access control rules, regular audits, and necessary updates to infrastructure.
- Ensures the overall security framework is followed organization-wide.

MULTIPURPOSE INTERNET MAIL EXTENSIONS

MIME (Multipurpose Internet Mail Extensions)

- **MIME** is the standard that lets email carry **more than plain text**. It defines how emails can include **attachments, images, audio, video, and different text formats** in a way that all mail servers and clients understand.
- Before MIME, email (SMTP era) could reliably carry **only 7-bit plain ASCII text**. No files, no images, no rich text.

The Problem MIME Solved

- Email systems were built for simple text. Binary files (PDF, JPG, MP3, DOCX) would get corrupted if sent directly.
- Different systems needed a **common labeling system** to know *what* the content is and *how* to read it.
- **MIME added labels + encoding rules** so any email app can correctly interpret the content.

MIME- HEADER FILE

```
MIME-Version: 1.0
Content-Type: multipart/mixed; boundary=frontier

This is a message with multiple parts in MIME format.
--frontier
Content-Type: text/plain

This is the body of the message.
--frontier
Content-Type: application/octet-stream
Content-Transfer-Encoding: base64

PGh0bWw+CiAgPGhlYWQ+CiAgPC9oZWFKPgogIDxib2R5PgogICAgPHA+VGhpcyBpcy8
Ym9keSBvZiB0aGUgbWVzc2FnZS48L3A+CiAgPC9ib2R5Pgo8L2h0bWw+Cg==
--frontier--
```

MIME-Version: 1.0: This header indicates that the message follows the MIME standard.

Content-Type: multipart/mixed: This tells the receiving email client that the message contains multiple parts (e.g., text and an attachment) that should be processed together.

Boundary (boundary=frontier): This acts as a "marker" to separate the different sections of the email. In the image, --frontier signals the start of a new part, and --frontier-- signifies the end of the entire message.

Content-Type: text/plain: The first part of the message is simple, unformatted text.

Content-Type: application/octet-stream: The second part is a binary file (an attachment).

Content-Transfer-Encoding: base64: Because email systems were originally designed for text, binary data (like images or PDFs) must be encoded into a text-safe format like Base64 to prevent corruption during transmission.

I. HOW MIME WORKS (STEP BY STEP)

Step 1: User creates an email

- You write: Text message and Attach a PDF/image/video. But, SMTP (basic email system) only understands **plain text**.

Step 2: MIME converts everything into a structured format

(A) Identifies content type

- It adds headers like: text/plain, text/html, image/jpeg, application/pdf
- Example: Content-Type: application/pdf

(B) Converts binary data into text (Encoding)

- Files like images or PDFs are binary (0s and 1s) so MIME converts them into **text using Base64 encoding**.
- Example: JVBERi0xLjQKJcfs... This ensures email systems can safely transmit it.

Step 3: Email is sent via SMTP

- Now everything is Converted into text and Properly labeled so that SMTP can send it like normal email data.

Step 4: Receiver's email client decodes MIME

- At the receiver side: Base64 → original file restored and Headers tell how to display it (image, PDF, HTML, etc.)

BASE-64 ENCODING

Index	Binary	Char	Index	Binary	Char	Index	Binary	Char	Index	Binary	Char
0	000000	A	16	010000	Q	32	100000	g	48	110000	w
1	000001	B	17	010001	R	33	100001	h	49	110001	x
2	000010	C	18	010010	S	34	100010	i	50	110010	y
3	000011	D	19	010011	T	35	100011	j	51	110011	z
4	000100	E	20	010100	U	36	100100	k	52	110100	0
5	000101	F	21	010101	V	37	100101	l	53	110101	1
6	000110	G	22	010110	W	38	100110	m	54	110110	2
7	000111	H	23	010111	X	39	100111	n	55	110111	3
8	001000	I	24	011000	Y	40	101000	o	56	111000	4
9	001001	J	25	011001	Z	41	101001	p	57	111001	5
10	001010	K	26	011010	a	42	101010	q	58	111010	6
11	001011	L	27	011011	b	43	101011	r	59	111011	7
12	001100	M	28	011100	c	44	101100	s	60	111100	8
13	001101	N	29	011101	d	45	101101	t	61	111101	9
14	001110	O	30	011110	e	46	101110	u	62	111110	+
15	001111	P	31	011111	f	47	101111	v	63	111111	/

S/MIME

- S/MIME (Secure/Multipurpose Internet Mail Extensions) is an extension of MIME that provides security features for email communication.
- It supports encryption for confidentiality, digital signatures for authentication, and message integrity.
- S/MIME ensures that email messages are protected from unauthorized access and tampering.

TYPES OF S/MIME MESSAGES

1. Enveloped Data

- Enveloped data provides confidentiality by encrypting the email message. A random session key is used to encrypt the message using AES, and the session key is encrypted using the receiver's RSA public key.

2. Signed Data

- Signed data ensures authentication and integrity. A message digest is generated using SHA-256 and then encrypted with the sender's private key to form a digital signature.

3. Clear-Signed Data

- Clear-signed data allows the message to remain readable while only the digital signature is encoded using Base64. This allows non-S/MIME users to read the message but not verify the signature.

4. Signed and Enveloped Data

- This type combines both signing and encryption. The message is first signed and then encrypted, providing the highest level of security.

HOW S/MIME WORKS?

- i. Sender writes email
- ii. Email is processed using MIME (for formatting + attachments)
- iii. Then S/MIME:
 - Encrypts the message using recipient's public key
 - Optionally signs it using sender's private key
- iv. Recipient decrypts using private key

PRETTY GOOD PRIVACY (PGP)

- PGP is an **encryption system** designed to:
- Protect emails and files from being read by unauthorized people
- Verify the identity of the sender (authentication)
- Ensure messages haven't been altered (integrity)

How it works

- PGP uses **public-key cryptography**:
- You have a **public key** (shared with others) and a **private key** (kept secret)
- If someone wants to send you a secure message, They encrypt it using your **public key**
- Only your **private key** can decrypt it

Common uses

- Secure email communication
- Protecting sensitive files
- Signing software releases
- Privacy-focused messaging

SECURE SOCKETS LAYER (SSL)

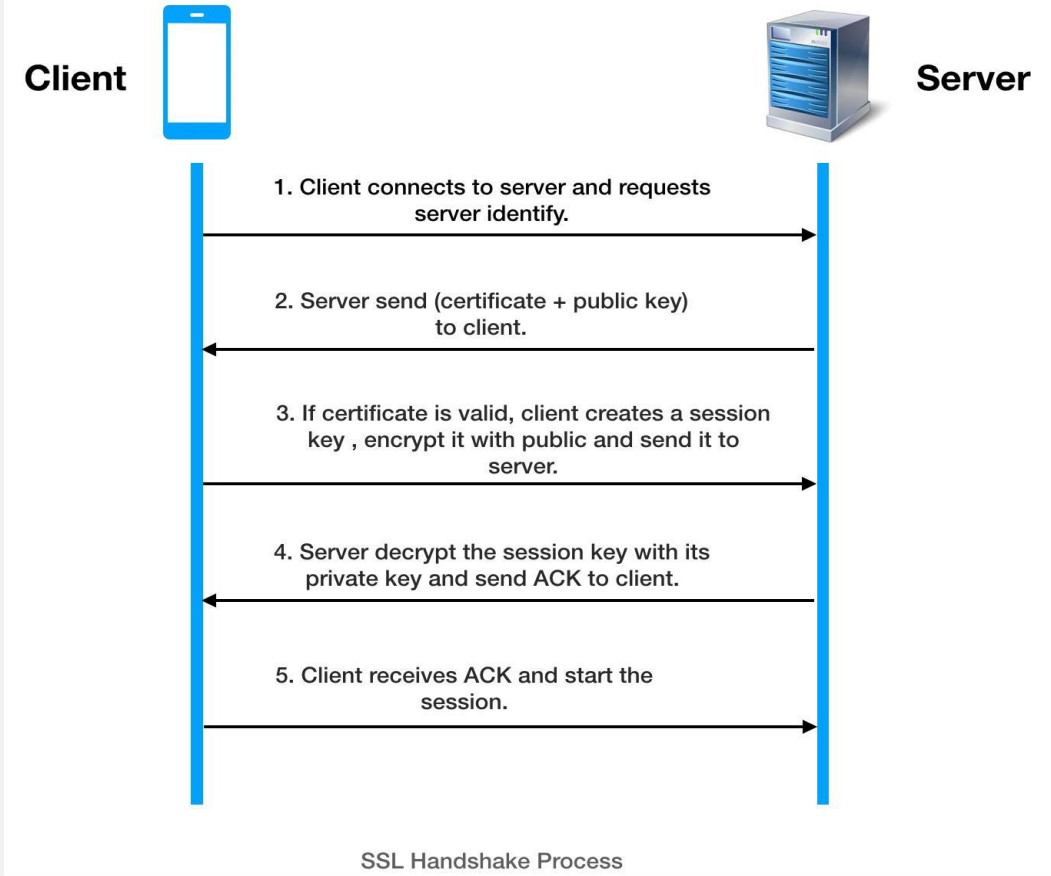
- **Secure Sockets Layer (SSL)** is a security protocol used to establish an encrypted and secure connection between a client (browser/user) and a server (website).
- Today, SSL has been replaced by the more secure Transport Layer Security (TLS), but people still commonly use the term “SSL”.
- Websites using SSL/TLS show **HTTPS** instead of HTTP in the URL, meaning the communication is encrypted and secure.

IMPORTANCE OF SSL

- Before SSL, web data was transmitted in plaintext, making it vulnerable to interception. SSL solves this by:
- Encrypting sensitive information such as login credentials, financial transactions and personal data.
- Authenticating web servers to prevent users from connecting to fraudulent websites.
- Ensuring data integrity so transmitted information cannot be modified during transit.

HOW THEY WORK

- **Client Hello** – Your browser sends a request to the server
- **Server Hello** – Server responds with its digital certificate
- **Authentication** – Browser verifies the certificate (issued by a trusted authority)
- **Key Exchange** – Both agree on encryption keys
- **Secure Communication** – Data is encrypted and safely transmitted



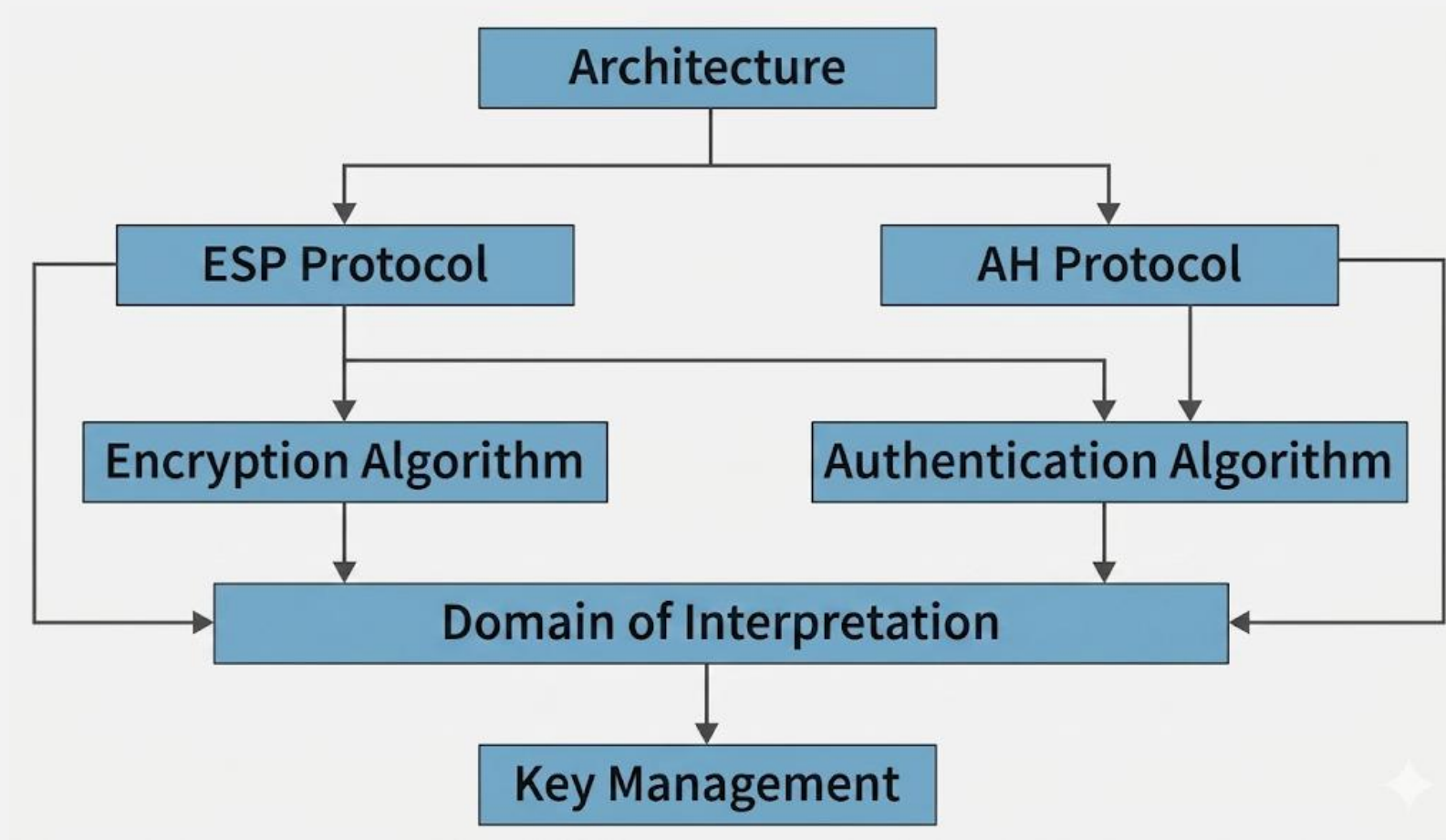
IPSEC (INTERNET PROTOCOL SECURITY)

- IPsec is a collection of protocols used to provide security for communication over IP networks such as the Internet.
- It protects data while it travels between devices by providing encryption, authentication, and integrity checking. IPsec mainly operates at the Network Layer (Layer 3) of the OSI model.

Features of IPsec

- It ensures **confidentiality** by encrypting data so that unauthorized users cannot read it.
- It provides **authentication** by verifying the identity of communicating devices.
- It maintains **data integrity** by ensuring that the data is not modified during transmission.
- It offers **anti-replay protection** by preventing attackers from resending captured packets.

COMPONENTS OF IPSEC



COMPONENTS OF IPSEC

1. Authentication Header (AH)

- The Authentication Header protocol provides authentication, integrity, and anti-replay protection for IP packets.
- It verifies that the data has not been altered during transmission and confirms the identity of the sender. However, AH does not encrypt the data, so the information remains visible.

2. Encapsulating Security Payload (ESP)

- The Encapsulating Security Payload protocol provides encryption along with authentication and integrity. ESP protects the confidentiality of data by encrypting it before transmission.
- Because it provides stronger security features, ESP is more commonly used than AH.

IPSEC CONNECTION ESTABLISHMENT PROCESS

Phase 1: IKE Tunnel Establishment

- In Phase 1, a secure channel is created using the Internet Key Exchange (IKE) protocol. This channel is used to securely negotiate security settings and exchange cryptographic keys for further communication.

Phase 2: IPsec Tunnel Establishment (Quick Mode)

- In Phase 2, the peers negotiate the IPsec Security Associations (SAs). These SAs define how the actual data will be protected during transmission.

FIREWALL

- A firewall is a security system designed to prevent unauthorized access to or from a private network.
- A firewall can be implemented using hardware, software, or a combination of both.
- All data entering or leaving the network passes through the firewall, which examines the traffic and blocks suspicious activity.
- There are several types of firewall techniques used to protect networks from unauthorized access. The main types include:
 - i. Packet filtering firewalls
 - ii. Application Gateways
 - iii. Circuit Level Gateways
 - iv. Stateful inspection firewalls.

PACKET FILTERING FIREWALL

- A packet filtering firewall operates at the network layer of the OSI model and is commonly implemented in routers.
- The firewall examines packets based on information such as source IP address, destination IP address, port numbers, and protocol type.
- Based on predefined rules, the firewall either forwards the packet or discards it.

Filtering Rules

- Packet filtering rules are defined by network administrators according to the organization's security requirements.
- Rules may allow traffic from trusted IP addresses while blocking traffic from unknown or suspicious sources.
- Rules can also restrict access to certain services by controlling port numbers, such as blocking access to specific applications or servers.

APPLICATION GATEWAY

- An application gateway is a type of firewall that provides security by controlling traffic at the application layer of the OSI model.
- It is also known as an application proxy or application-level proxy because it acts as an intermediary between clients and servers.
- Instead of allowing direct communication between the internal network and external network, the application gateway processes all requests on behalf of the client.
- An application gateway can inspect the actual content of messages rather than only examining packet headers.
- It can enforce authentication mechanisms such as usernames and passwords before allowing access to resources.
- It can log user activities and monitor traffic for suspicious behavior.

CIRCUIT-LEVEL GATEWAY

- A circuit-level gateway is a type of firewall that monitors and controls network traffic at the session layer of the OSI model.
- It does not inspect the actual contents of packets but instead verifies whether the connection between two systems is legitimate.
- The main purpose of a circuit-level gateway is to ensure that a valid session is established before allowing data transmission.

STATEFUL INSPECTION FIREWALL

- A stateful inspection firewall is a type of firewall that monitors the state of active connections and makes decisions based on the context of the traffic.
- Unlike packet filtering firewalls, it examines both packet headers and the state of the connection to determine whether packets are allowed to pass.
- This type of firewall operates at multiple layers of the OSI model, typically the network and transport layers.

How Stateful Inspection Works

- The firewall keeps track of the state of each connection in a state table, including information such as source and destination IP addresses, ports, and sequence numbers.
- It verifies that incoming packets correspond to an existing valid connection before allowing them to enter the network.
- If a packet does not match any active connection or appears to be malicious, the firewall blocks it.
- This approach allows the firewall to detect unauthorized or unexpected traffic that would bypass simple packet filters.